

Lecture 6: Estimating the variance of errors

Economics 326 — Introduction to Econometrics II

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The importance of σ^2

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We use the notation $E[\cdot | \mathbf{X}] = E[\cdot | X_1, \dots, X_n]$.

- The variance of $\hat{\beta}$ depends on the unknown $\sigma^2 = E[U_i^2]$:

$$\text{Var}(\hat{\beta} | \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- If U 's were observable, we could estimate σ^2 by $\frac{1}{n} \sum_{i=1}^n U_i^2$, which is unbiased, but infeasible.
- Using residuals $\hat{U}_i = Y_i - \hat{\alpha} - \hat{\beta}X_i$ gives a feasible estimator

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \hat{U}_i^2,$$

but $\hat{\sigma}^2$ is biased.

An unbiased estimator of σ^2

- An unbiased estimator of σ^2 is

$$s^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{U}_i^2.$$

- Assumptions:
 1. $Y_i = \alpha + \beta X_i + U_i$.
 2. $E[U_i | \mathbf{X}] = 0$ for all i .
 3. $E[U_i^2 | \mathbf{X}] = \sigma^2$ for all i .
 4. $E[U_i U_j | \mathbf{X}] = 0$ for all $i \neq j$.
- Since $\hat{U}_i = Y_i - \hat{\alpha} - \hat{\beta}X_i$, dividing by $n-2$ adjusts for estimating α and β .

Expressing $\hat{U}_i$ in terms of U_i

From

$$\begin{aligned}\hat{U}_i &= (Y_i - \bar{Y}) - \hat{\beta}(X_i - \bar{X}), \\ Y_i - \bar{Y} &= \beta(X_i - \bar{X}) + U_i - \bar{U},\end{aligned}$$

we get

$$\hat{U}_i = (U_i - \bar{U}) - (\hat{\beta} - \beta)(X_i - \bar{X}).$$

Expanding $\sum_{i=1}^n \hat{U}_i^2$

$$\hat{U}_i^2 = (U_i - \bar{U})^2 + (\hat{\beta} - \beta)^2 (X_i - \bar{X})^2 - 2(\hat{\beta} - \beta)(X_i - \bar{X})(U_i - \bar{U}).$$

Thus,

$$\sum_{i=1}^n \hat{U}_i^2 = \sum_{i=1}^n (U_i - \bar{U})^2 + (\hat{\beta} - \beta)^2 \sum_{i=1}^n (X_i - \bar{X})^2 - 2(\hat{\beta} - \beta) \sum_{i=1}^n (X_i - \bar{X})(U_i - \bar{U}).$$

To show $E[\sum_{i=1}^n \hat{U}_i^2] = (n-2)\sigma^2$, we verify the three expectations below.

Expectation of $\sum_{i=1}^n (U_i - \bar{U})^2$

$$\begin{aligned} \sum_{i=1}^n (U_i - \bar{U})^2 &= \sum_{i=1}^n U_i^2 - \frac{1}{n} \left(\sum_{i=1}^n U_i \right)^2 \\ &= \sum_{i=1}^n U_i^2 - \frac{1}{n} \left(\sum_{i=1}^n U_i^2 + \sum_{i=1}^n \sum_{j \neq i} U_i U_j \right). \end{aligned}$$

Taking expectations and using the assumptions,

$$E \left[\sum_{i=1}^n (U_i - \bar{U})^2 \right] = n\sigma^2 - \frac{1}{n} n\sigma^2 = (n-1)\sigma^2.$$

Expectation of $(\hat{\beta} - \beta)^2 \sum_{i=1}^n (X_i - \bar{X})^2$

Because $E[\hat{\beta}] = \beta$ (conditionally on $\mathbf{X}$),

$$E[(\hat{\beta} - \beta)^2] = \text{Var}(\hat{\beta}) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

Hence,

$$E \left[(\hat{\beta} - \beta)^2 \sum_{i=1}^n (X_i - \bar{X})^2 \right] = \sigma^2.$$

Expectation of $(\hat{\beta} - \beta) \sum_{i=1}^n (X_i - \bar{X})(U_i - \bar{U})$

Note that

$$\sum_{i=1}^n (X_i - \bar{X})(U_i - \bar{U}) = \sum_{i=1}^n (X_i - \bar{X})U_i,$$

and

$$\hat{\beta} - \beta = \frac{\sum_{i=1}^n (X_i - \bar{X})U_i}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

Therefore,

$$(\hat{\beta} - \beta) \sum_{i=1}^n (X_i - \bar{X})(U_i - \bar{U}) = \frac{1}{\sum_{i=1}^n (X_i - \bar{X})^2} \left(\sum_{i=1}^n (X_i - \bar{X})U_i \right)^2.$$

Conditionally on $\mathbf{X}$,

$$E \left[(\hat{\beta} - \beta) \sum_{i=1}^n (X_i - \bar{X})(U_i - \bar{U}) \right] = \frac{1}{\sum_{i=1}^n (X_i - \bar{X})^2} \left(\sigma^2 \sum_{i=1}^n (X_i - \bar{X})^2 \right) = \sigma^2.$$

Putting it together

Using the three expectations above,

$$\mathbb{E} \left[\sum_{i=1}^n \hat{U}_i^2 \right] = (n-1) \sigma^2 + \sigma^2 - 2\sigma^2 = (n-2) \sigma^2,$$

so s^2 is unbiased for σ^2 .

Estimating the variance of $\hat{\beta}$

- Variance of $\hat{\beta}$ (conditional on $\mathbf{X}$):

$$\text{Var}(\hat{\beta}) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- Estimator of σ^2 :

$$s^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{U}_i^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta}X_i)^2.$$

- Estimator of the variance of $\hat{\beta}$:

$$\widehat{\text{Var}}(\hat{\beta}) = \frac{s^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- Standard error of $\hat{\beta}$:

$$\text{SE}(\hat{\beta}) = \sqrt{\widehat{\text{Var}}(\hat{\beta})} = \sqrt{\frac{s^2}{\sum_{i=1}^n (X_i - \bar{X})^2}}.$$